NILAY KUMAR

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PROFESSIONAL SUMMARY

Third-year B.Tech CSE student at Delhi University (2024–2028) with hands-on experience building full-stack AI applications across healthcare, agriculture, and enterprise domains. Developed and shipped three end-to-end projects integrating React, Firebase, and Google Gemini AI — covering everything from system architecture to deployment. Passionate about building accessible, real-world technology that bridges the gap between cutting-edge AI and meaningful user problems.

CORE SKILLS

Python & C	Data Structures	Generative AI	Google Gemini AI	GitHub	Data Analysis
Java Script	Machine Learning	Full Stack Dev	Firestore, Auth, Cloud Functions	React.js, Tailwind CSS	Database Firebase Firestore

EXPERIENCE

Build With Gemini Hackathon | Design Innovation Centre, University of Delhi *(Secured 3rd Place) 21 Nov 2025*

- Secured **3rd place** at the Hackathon, competing among 60+ teams in the finale and presenting an AI-driven mental health solution.
- Collaborated in a fast-paced team environment to refine ideas, solve real-world problems, and deliver a functional prototype under strict timelines.
- Incorporated expert feedback from judges to enhance system scalability, technical depth, and real-world applicability of the solution.

Artificial Intelligence Intern | CodeAlpha *Jan 2025 – Feb 2025*

- Developed a Language Translation Tool using external APIs, enabling seamless multi-language text conversion.
- Built an FAQ Chatbot leveraging NLP libraries (NLTK/SpaCy) to deliver instant, accurate responses to user queries.
- Delivered two production-ready AI projects within a one-month timeline.

Publications & Research

DocuMind: AI-Powered BRD Generation System | *IRAI 2026 (IEEE)* IEEE International Conference on Responsible Artificial Intelligence · Melbourne, Australia

- Engineered and Architected an end-to-end RAG pipeline ingesting unstructured documents, classifying snippets via Gemini-powered semantic labeling, and ranking embeddings using cosine similarity over Firestore vector store.
- Research paper accepted and presented at IEEE IRAI 2026, selected from 575 submissions across 57 countries.
- Designed a BRD Synthesis Engine using Gemini 2.5 Flash generating 9 structured sections simultaneously with inline source citations, backed by a 3-dimension Quality Scoring Algorithm and an AI Quality Auditor for real-time gap detection and document patching

FEATURED PROJECTS

DocuMind – AI BRD Generation Agent

- Engineered a multi-source ingestion pipeline (Slack, Gmail, PDFs, transcripts) with a hybrid retrieval system combining BM25, vector search (ChromaDB), and cross-encoder reranking to generate structured 10-section BRDs in under 30 seconds
- Implemented 100% sentence-level traceability linking every generated requirement back to its original source, alongside graph-based conflict detection with 95%+ accuracy
- Built a real-time AI Quality Auditor scoring BRDs on completeness, clarity, and consistency, with auto-generated PlantUML use case diagrams and one-click PDF export

FasalSetu – Smart Agriculture & AI Crop Advisory System

- Developed an AI-powered agri-tech platform integrating smart soil sensors (NPK, moisture) with a conversational chatbot to deliver real-time, personalized crop guidance to farmers.
- Designed a hardware-software system for pest/disease detection, automated alerts, and an in-house smart plant-growing kit enabling supplementary farmer income
- Built a community-impact-first business model combining hardware kit sales, AI advisory subscriptions, and agri-platform partnerships targeting rural digital adoption

MannMitra – AI-Powered Mental Health Platform

- Built a full-stack mental wellness platform with AI conversation, voice therapy, and camera-based real-time emotion detection with cultural sensitivity adaptation.
- Architected a multi-service backend using Google Gemini/Vertex AI, Firebase Firestore, and a crisis detection engine with risk assessment and emergency escalation protocols
- Implemented end-to-end security (JWT, Firebase Auth, local-first emotion processing) with GDPR compliance and PWA support.

EDUCATION

Delhi University – Faculty of Technology	B.Tech – Computer Science – 9.05 CGPA	*2024 – 2028*
Delhi Public School	Higher Secondary Certificate – 92%	*2022 – 2024*
Ryan International School	Secondary School Certificate – 96.8%	*2014 – 2022*

CERTIFICATIONS & COURSES

Introduction to Artificial Intelligence – IBM	Generative AI – Certified
Machine Learning for All – University of London	Introduction to Game Development using Scratch – Coursera
Natural Language Processing (NLP) – Certified	Introduction to Git and GitHub – Google